Proposal for a Bring-Your-Own-Device Requirement for Computer Science Majors

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April 5, 2026

Abstract

Starting in 2027–28, the Department of Computer Science wishes to require majors and minors, at both undergraduate and graduate levels, to own a portable, current, and capable computing device that supports the software and technical demands of the curriculum. The policy defines minimum hardware and software standards, identifies unacceptable devices, and strongly encourages compliance for non-majors enrolled in selected CS courses built around regular in-class computer use.

The policy aims to ensure that all students have access to the necessary technology to fully engage with the coursework, which increasingly relies on hands-on programming, software development, and cybersecurity exercises. By setting clear guidelines for device specifications and usage, the department seeks to enhance the learning experience, promote equity in access to technology, and prepare students for the technical demands of their future careers in computer science.

Bring Your Own Device

Overview

Beginning with academic year 2027-28, undergraduate and graduate students enrolled in any major or minor offered by the Department of Computer Science are required to own and use a computer device that meets the specifications of this policy.

If you enroll in a Computer Science course, without being a CS major or minor, you are strongly encouraged to comply with this policy. The following courses are build around the assumption that every student in the classroom has a qualified device as defined in this policy: COMP 125, COMP 141, COMP 150, COMP 170, COMP 264, COMP 271, COMP 272, and COMP 363.

Qualified Devices

A *qualified device* is a system that meets the expectations of this policy, by featuring all of the following minimum requirements.

Minimum Requirements

Screen. The device's screen must have at least a 10-inch diagonal, a 16:9 aspect ratio (or comparable aspect ratio), and at least 1204×768 pixels resolution.

Keyboard. The device must have a separate physical keyboard.

Memory. At least 16 GB of memory.

Storage. At least 128 GB of storage.

Neworking. WiFi capable, compatible with IEEE 802.11ac (also known as WiFi-5) or higher.

Battery life. A fully charged system should be able to run for at least 6-7 hours.

Tracking. The device must be trackable so that it can be located if forgotten/stolen.

No older than 4 years.

Portable. The system must be light enough to be carried around campus comfortably and not require excessive space on a classroom desk.

Operating system. A qualified device must be capable of running the current version of Microsoft Windows, MacOS, Chromebook OS, or a major Linux distribution, specifically Ubuntu or Kali.

Software. The system must be capable of running the current version of the following applications: any major browser (Google Chrome, Microsoft Edge, or Apple Safari). It must also support native or in-browser versions of Microsoft Teams, Microsoft Office (at least Outlook, Word and Powerpoint), and Adobe Acrobat. Furthermore, the system must be capable of supporting these applications and their future updates for the duration of your studies at LUC, which means that the system cannot be more than 3-4 years old at the time you join Loyola University Chicago.

Access to Command Line Interface. The system must have a feature that allows direct access to the command line interface. For example, the Terminal application in MacOS, Windows, and Linux machines.

Optional Requirements

Expandable. The system should be able to connect to various peripherals such as a larger screens (if you have one in your residence), printers, etc.

Protection. The device must be covered by a protection policy (an insurance policy to cover serious repairs and provide replacement in case of major damage).

Terms of use

1. Your system must be password protected.

2. Certain courses in your major will be marked as "device required in classroom". For those courses you are expected to arrive to the classroom with your device. The battery charge in the device should be sufficient to last for the duration of the class meeting.

Absolutely not acceptable systems

Following is a list of systems and devices that do not meet the requirements.

1. E-ink devices (e.g., reMarkable)
2. Smart phones. Typically their screens are smaller than the required minimum size.
3. Desktop systems. They are not portable.
4. Chromebooks. They do not meet the software requirements for some cybersecurity courses.
5. Older systems. Such computers may not be equipped to run the latest version of software needed for the classes.
6. Older operating systems: Windows 10 or lower, MacOS 13 or lower, and Linux distributions released before 2024. Such systems may not be equipped to run the latest version of software needed for the classes, and may have security vulnerabilities.

Frequently Asked Questions

Is the purchase of a qualified device covered by financial aid?
Yes.

Are tablets like iPad or Microsoft Surface allowed?
Yes, as long as they meet the minimun requirements in this policy and especially the screen size, keyboard, and device age specifications.

Are Chromebooks allowed?
No. Chromebooks do not meet the software requirements for some cybersecurity courses. If you never plan to take cybersecurity courses, you can use a Chromebook, but we recommend getting a more capable system that will allow you to take any course in the curriculum.

Dean’s Cabinet Q&A

In March 2026, the department received feedback from the CAS Dean Cabinet. This section addresses some/as many of the questions posed by the cabinet.

What is approximately the cost of needed software or licenses (if needed) and would any discounts be available, for example if 100 student utilize it can either software or hardware be discounted?

The software our students need is either directly free, free with an academic license, or free because it’s covered by Loyola already (eg, Office 365 and TechConnect).

Have you considered that charging towers might be a necessity for the students?

Yes, and that is one reason we recommend relatively recent laptops with battery life sufficient for a full day of classes, thereby minimizing or eliminating the need for charging stations. The installation of charging infrastructure, however, is beyond the department’s capacity. Equipping classrooms for contemporary instruction is ultimately an institutional responsibility and charging towers are the *least* of the challenges we face in the classroom for CS instruction. Ample desk surface is far more critical and absent from most classrooms in Mundelein Center or Dumbach Hall. Faculty have previously offered recommendations regarding classroom requirements for adequate Computer Science instruction, though those suggestions have not been acted upon.

Have you explored a grant covering this?

Yes, but at present that does not appear to be a viable path. In the past, vendors such as Dell sometimes offered substantial discounts to students through university partnerships, packaged as grants and tied to broader service or equipment contracts. That model seems far less common now, likely because margins are tighter and many institutions have shifted significant portions of IT support and infrastructure to SaaS-based providers. We are not aware of foundations or agencies that routinely fund student equipment needs, except for underserved populations and communities. We will explore potential grant opportunities brought to our attention, but we should not delay implementation of this policy in the hope that such funding may materialize.

Check with our school of business and with the school of nursing to see how they manage their programs that strongly recommend computers. Note in your proposal how this is handled.

While it may be useful to know what other units do, our decision should ultimately rest on the nature of computer science itself. Computing is not peripheral to our discipline; it is the discipline. For that reason, our expectations around student access to computing platforms are likely to be more central, and more academically justified, than in many other programs.

How will you handle this for students who show an economic need and cannot afford the laptop?

These cases should be addressed individually and with flexibility. Some students may be able to obtain suitable refurbished equipment at substantially lower cost through vendors such as Newegg or similar sources. In addition, we could explore establishing a modest departmental or institutional fund to support a small loaner pool of laptops for students with demonstrated financial need, modeled on the USF Laptop Program.

AJCU practices

The following is a list of AJCU members that require students to bring their own laptops. Links to their BYOD policies are provided.

- Creighton University
  https://www.creighton.edu/student-technology-requirements/.
- Gonzaga University
  https://www.gonzaga.edu/school-of-engineering-applied-science/about/laptop-requirements.
- Marquette University
  https://www.marquette.edu/engineering/coetech/laptop-requirements.php.
- Rockhurst University
  https://www.rockhurst.edu/college-business/laptop-requirements.
- Seattle University
  https://www.seattleu.edu/science-engineering/academic-departments/department-of-computer-science/laptop-requirements/.

- University of San Francisco
  https://www.usfca.edu/arts-sciences/computer-science/technology-requirements.